



OPERATION FIRE READY (OFR)



1504204 AR,VR And MR System Design

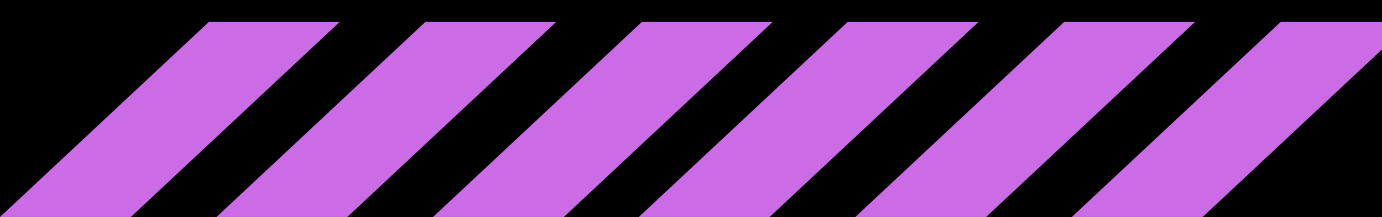
Group No. 12

Team Members

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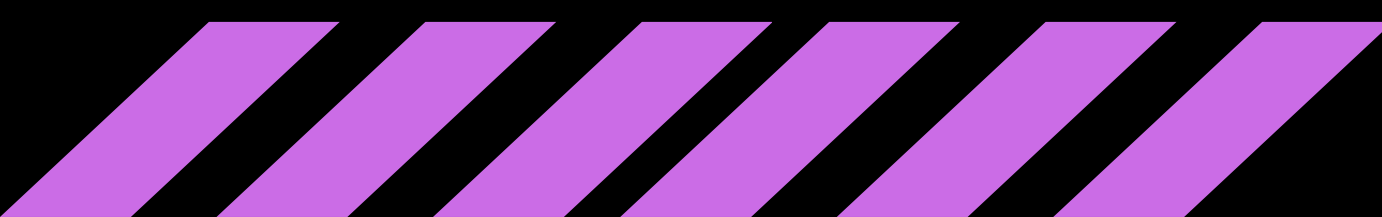
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Background for "Operation Fire Ready (OFR)"

"Operation Fire Ready (OFR)" is designed to address these gaps by leveraging advanced technologies and innovative training methods to create a more comprehensive, realistic, and flexible fire preparedness program.





Problem Identification



1

Lack of Real-World Stress in Drills: In traditional fire drills.

2

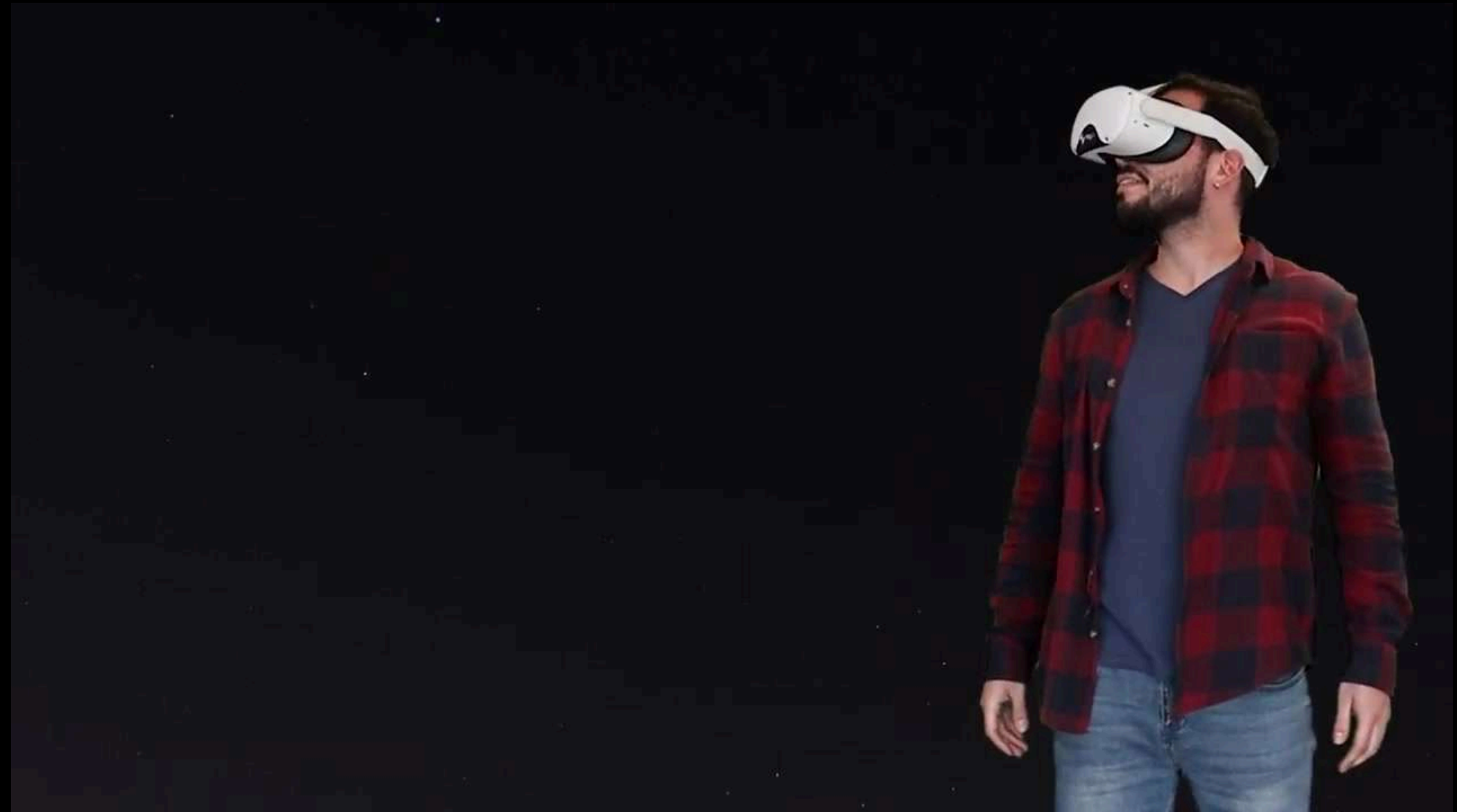
Difficulty in Training for High-Risk Environments: Conducting live fire drills in sensitive or high-risk environments, such as hospitals, industrial complexes, or densely populated areas.

3

Limited Scenarios and Flexibility: Conventional fire drills are often repetitive and limited to a few standardized scenarios, which may not represent the wide variety of realworld fire situations.



References





Objective

Create Realistic Training Scenarios: Develop immersive VR scenarios that replicate diverse fire situations, allowing users to practice decision-making and evacuation under realistic stress conditions.

Target Audience

Educational Institutions, Corporate Environments, Public Safety Organizations, Government and Municipalities, Vulnerable Populations.





Methodology



Needs Assessment (Week 1)

This stage involves a detailed analysis of the current gaps in fire safety training and drills across various sectors. It focuses on understanding the unique needs of different environments to create highly relevant fire scenarios.

VR Scenario Development (Week 2)

The second phase focuses on developing highly immersive and realistic VR fire scenarios using Unity. This involves close collaboration between developers, safety experts, and stakeholders.

IoT Integration (Week 3)

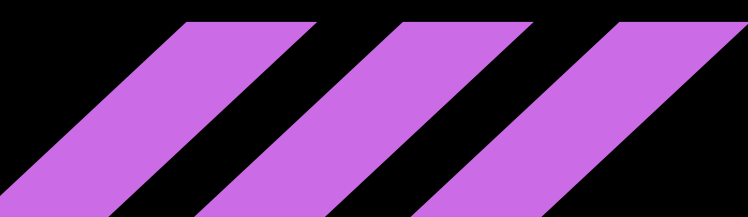
Integrating IoT sensors into the VR training system allows real-time environmental data to be used for simulation customization and enhanced realism.

Prototype Testing (Week 4)

This phase involves testing the developed VR prototype in a controlled environment to evaluate both the realism and the integration of IoT systems.

Training Program Deployment (Month 2 Onward)

After the development and testing phase, the program can be deployed across institutions and organizations.



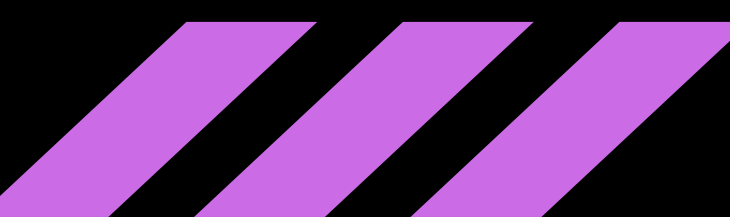
Key Features and Functionality

Users can navigate realistic simulations of various locations to experience fire emergencies in context. Diverse Scenario Options.



Technology and Tools Required

Unity, Oculus quest 2, Arduino board , Sensors , Empty fire extinguisher.

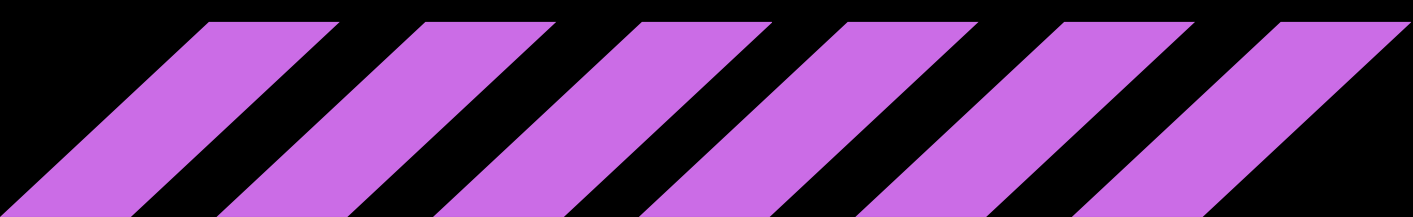




Expected Outcomes

Enhanced preparedness, improved engagement and retention, personalized feedback, faster response times, and comprehensive training materials.

These outcomes aim to create a safer environment and promote a culture of fire safety across various organizations and populations.

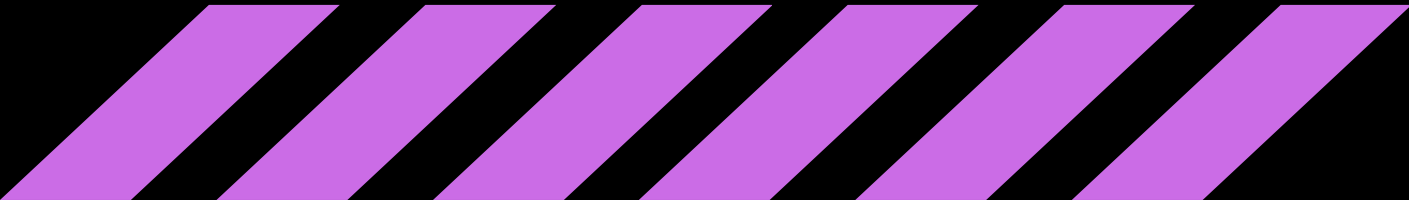


Timeline Table

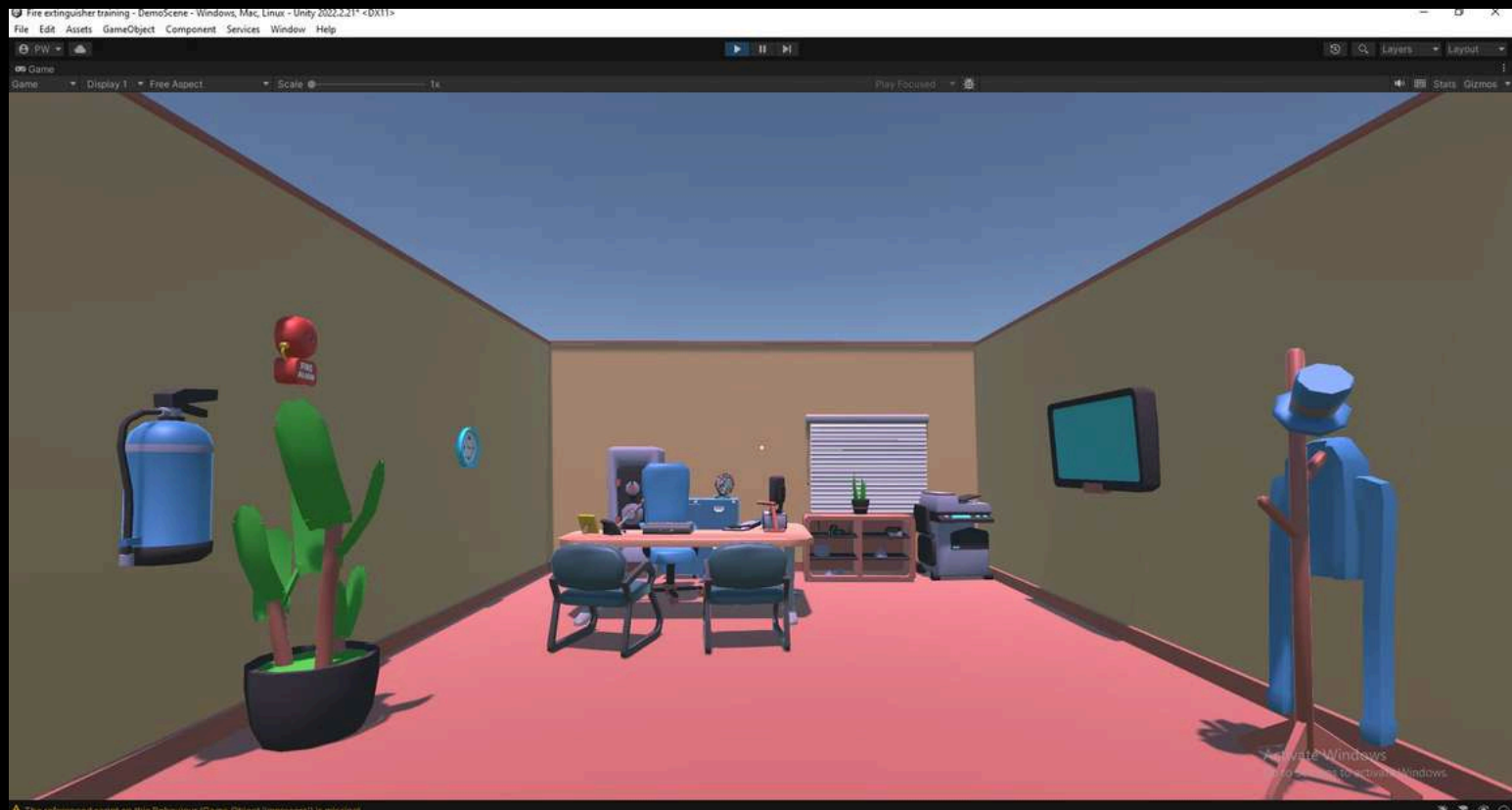


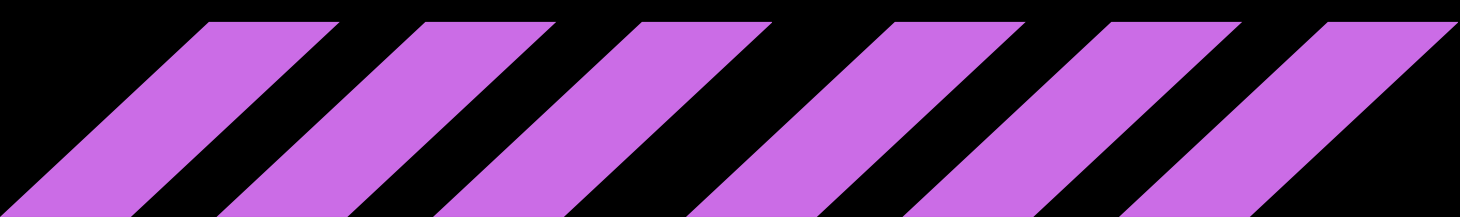
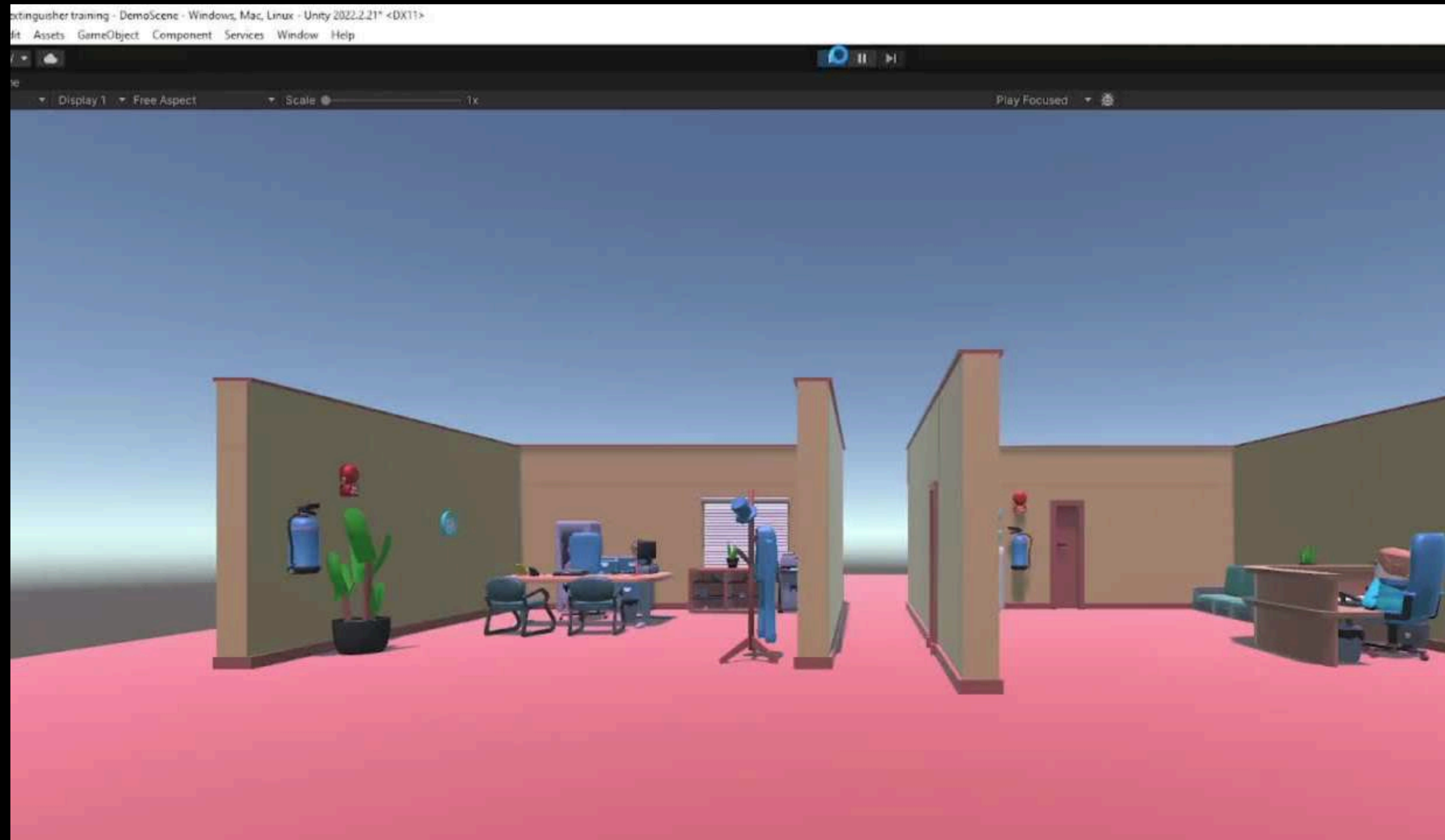
Month 1: Planning and Development

Week 1	Needs Assessment Conduct surveys: Gather insights from stakeholders on current fire drill practices and challenges. Identify key features: Define specific VR scenarios and IoT integrations needed for effective training.
Week 2	VR Content Development Develop VR scenarios: Create realistic firefighting drills that reflect different environments.
Week 3	IoT Setup Identify IoT devices: Select the relevant IoT devices. Develop an integration plan: Plan how IoT data will be collected and used during training.
Week 4	Technology Testing Prototype testing: Set up a prototype of a VR training program and test integration with IoT devices.



Week 11 Final Project







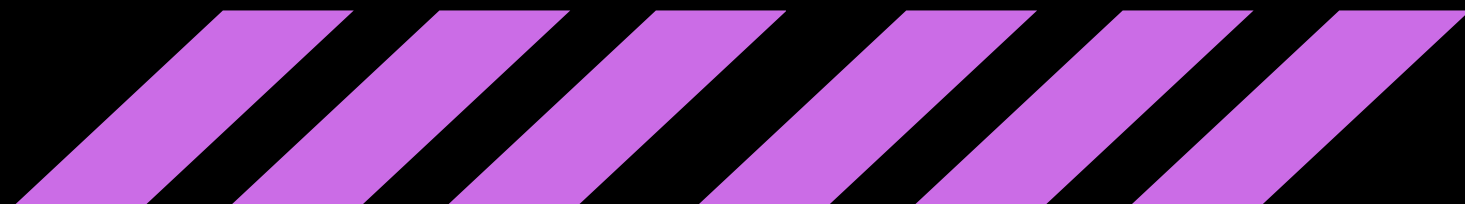
Week 12 Progress Final Project



Office room

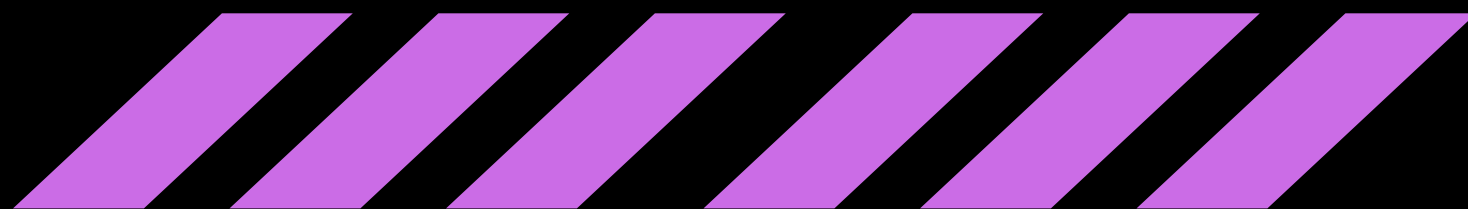
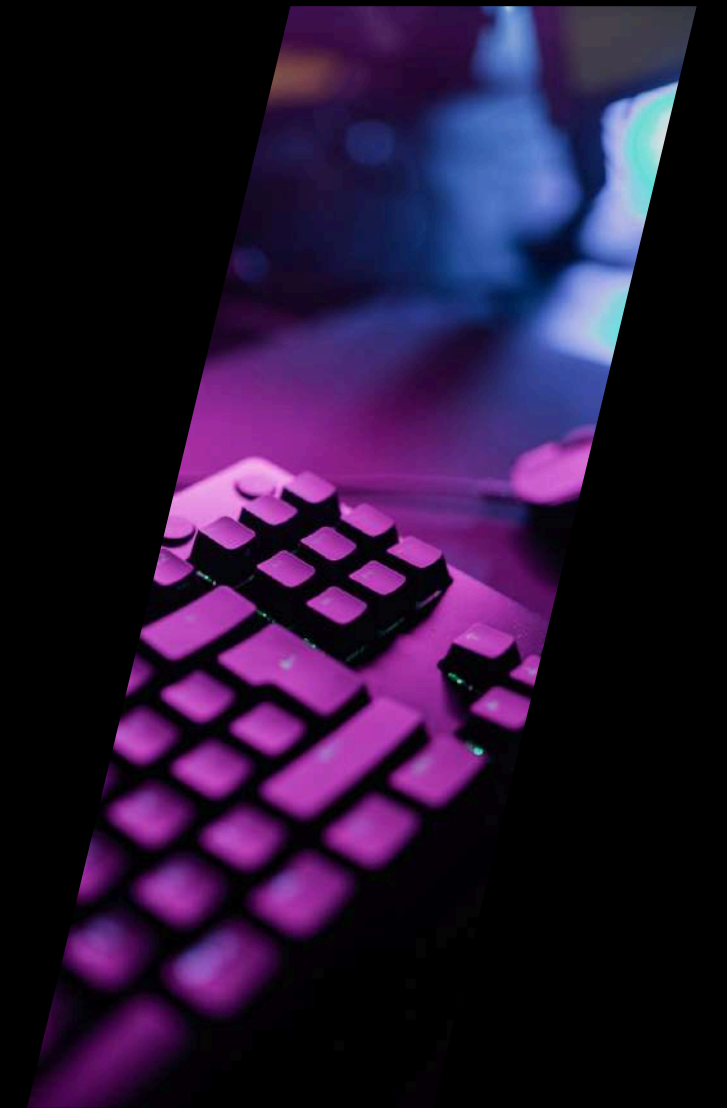


Generator room



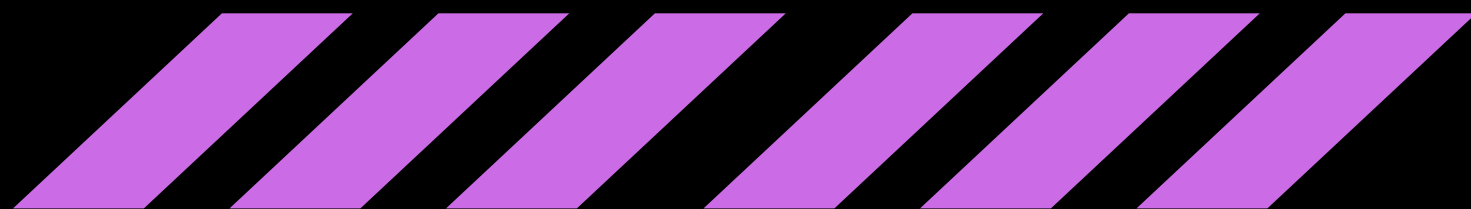


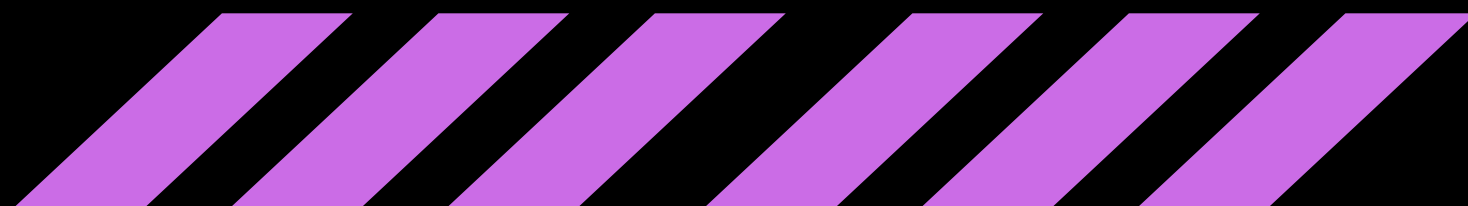
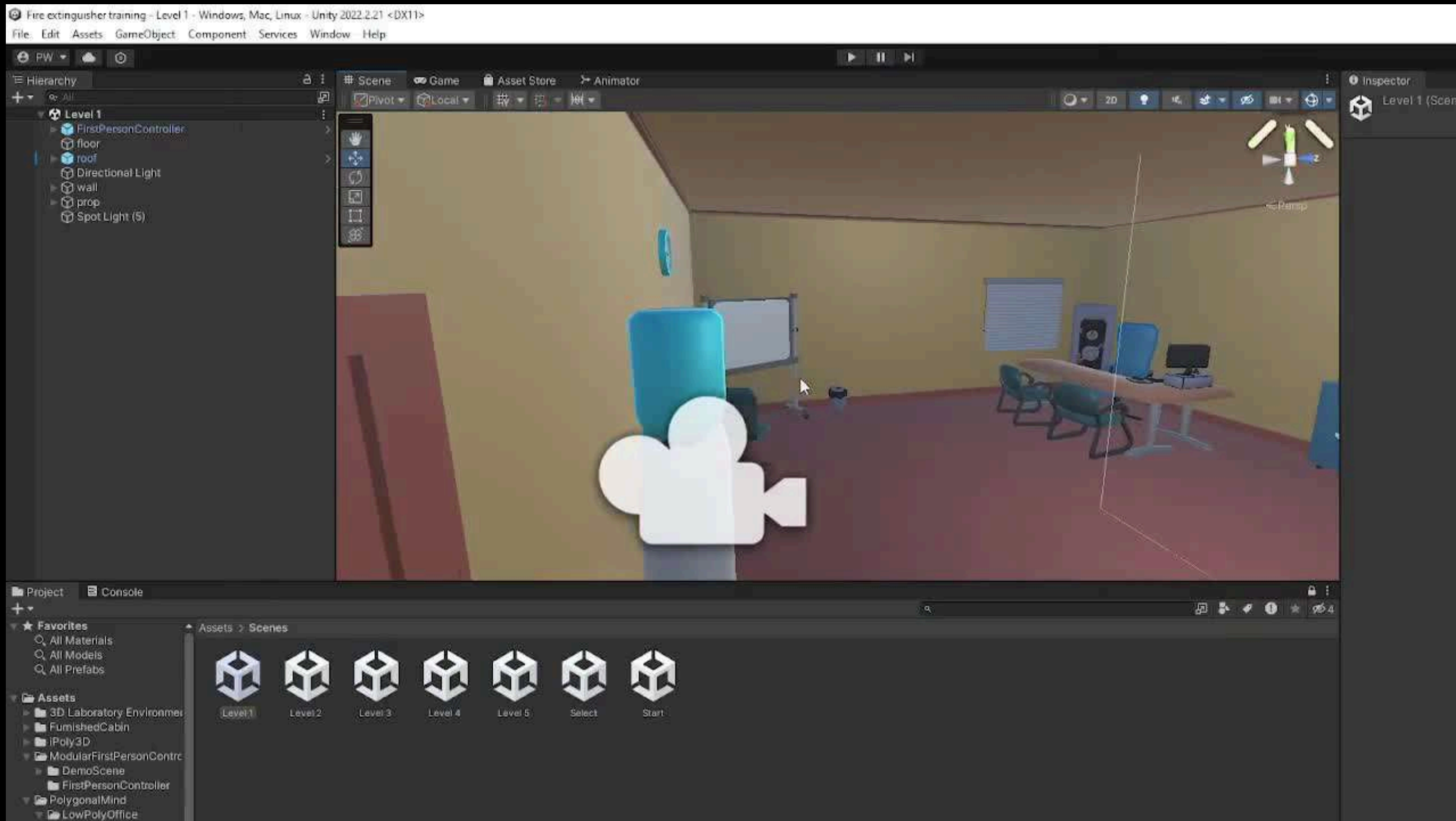
Chemistry room





Living room







Week 13 Progress Final Project



Operation Fire Ready
OFR



START

OPTION

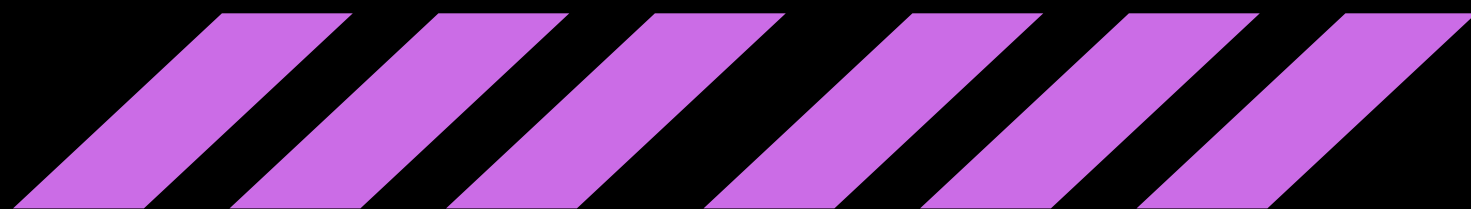
QUIT

UI Scene

OPTIONS



BACK

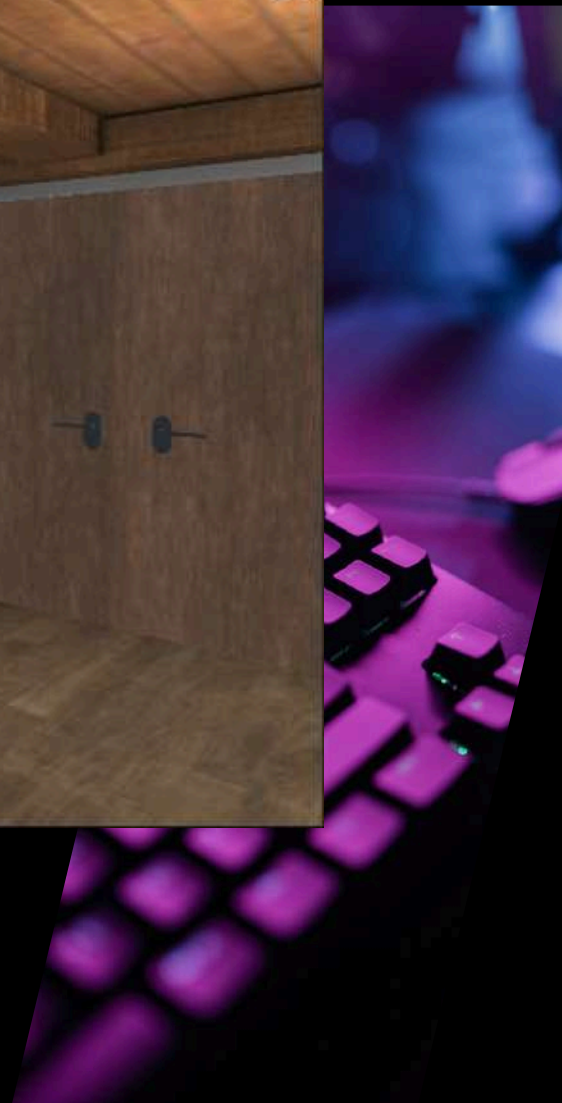
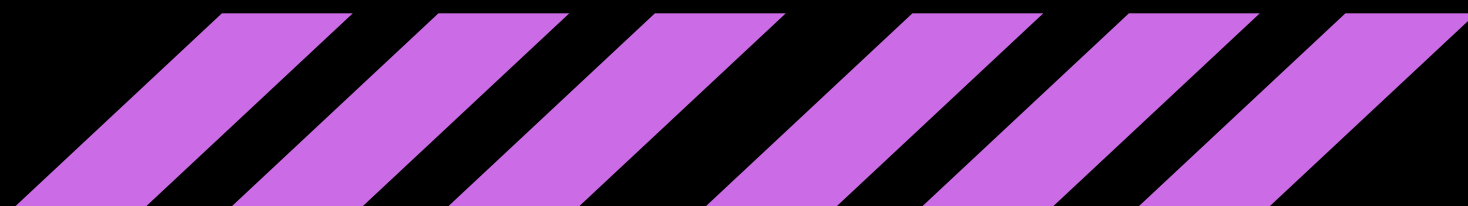


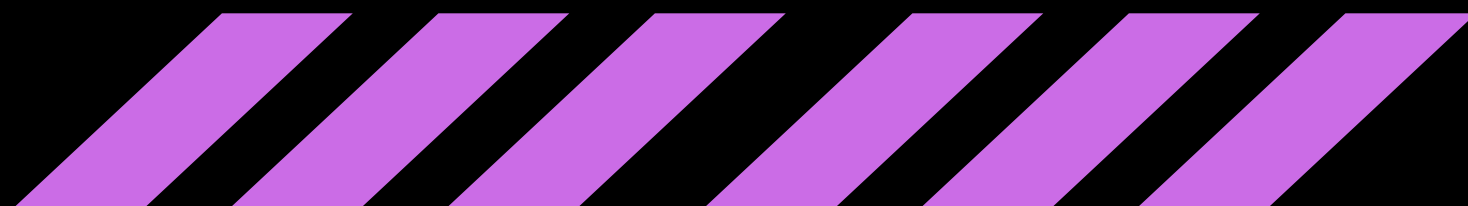


← **How to use**



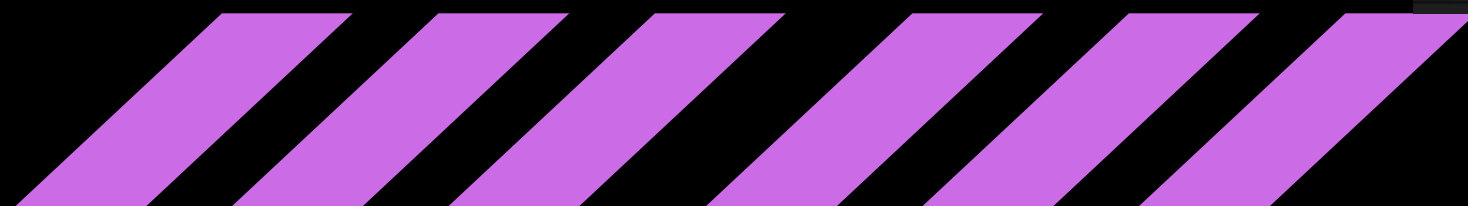
Fire in Living room →

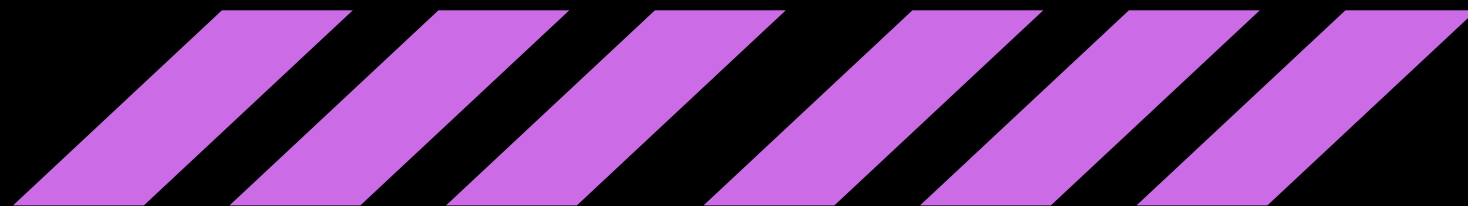


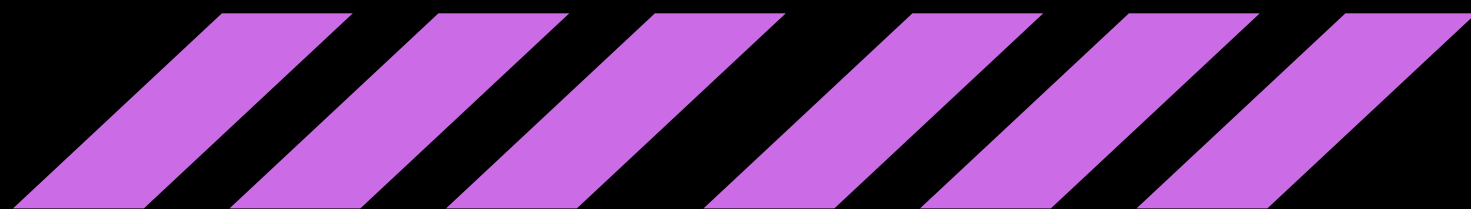
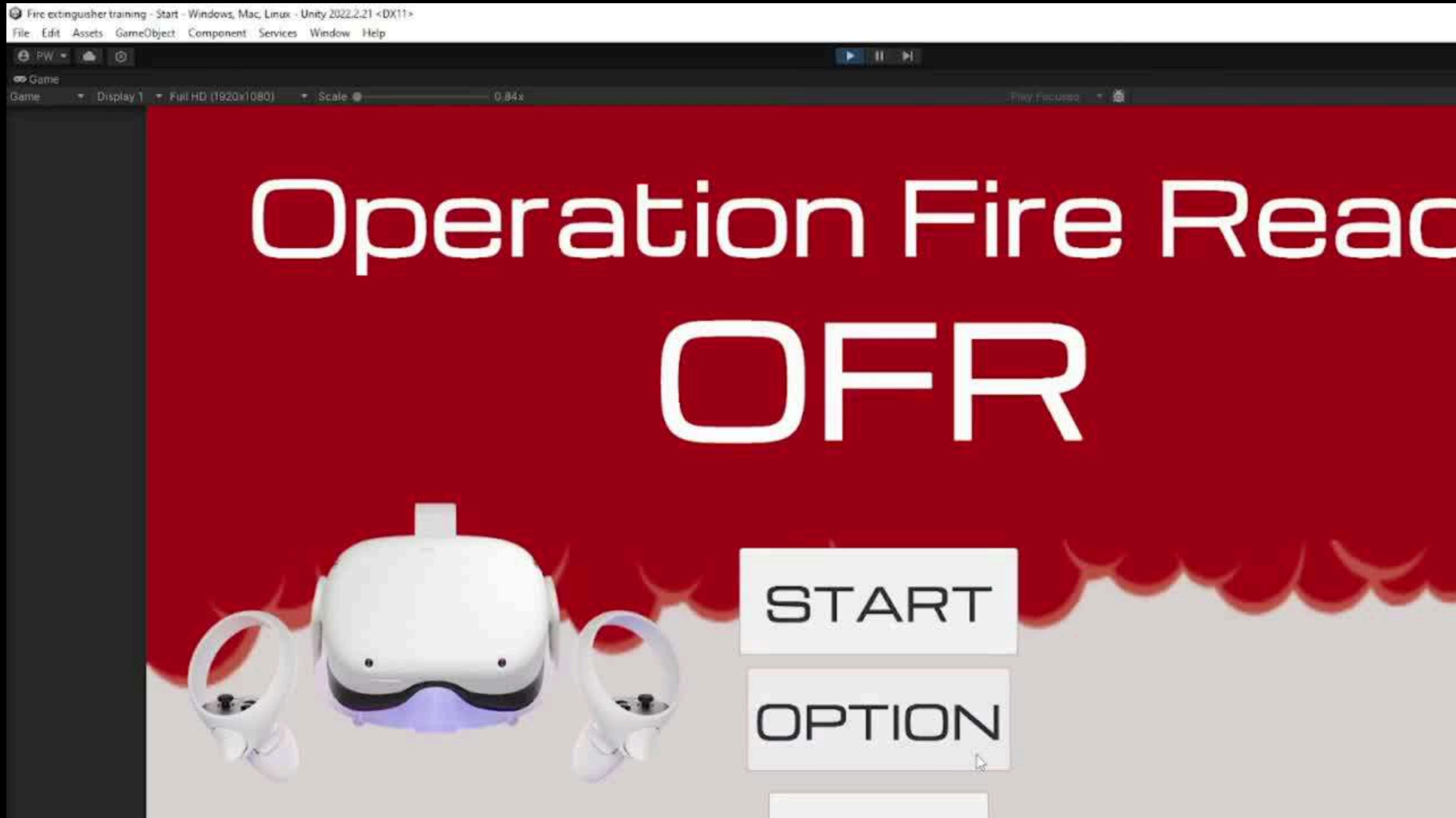




Week 14 Progress Final Project









Week 16 Final Project



Operation Fire Ready OFR



START

OPTION

QUIT





Thank You

